Question 1: Print Numbers from 1 to N

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Numbers_1toN_Using_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Number: ");
        int num=sc.nextInt();

        for(int i=1;i<=num;i++){
            System.out.print(i+" ");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Number: 10
1 2 3 4 5 6 7 8 9 10
Process finished with exit code 0
```

Question 2: Print Multiples of 3 between 1 and N

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Mul_3_1toN_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Number: ");
        int num=sc.nextInt();

        for(int i=1;i<=num;i++){
            int mul=3*i;
            if(mul<=num){
                System.out.print(mul+" ");
            }
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Number: 15
3 6 9 12 15
Process finished with exit code 0
```

Question 3: Calculate the Factorial of a Number

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Fact_of_num_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Number: ");
        int num=sc.nextInt();

        int mul=1;
        for(int i=1;i<=num;i++){
            mul=mul*i;
        }
        System.out.println("Factorial of "+num+" is "+mul);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Number: 6
Factorial of 6 is 720

Process finished with exit code 0
```

Question 4: Print Even Numbers from 1 to N

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Even_no_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num=sc.nextInt();

        for(int i=1;i<=num;i++){
            if(i%2==0){
                System.out.print(i+" ");
            }
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe" '
Enter a number: 10
2 4 6 8 10
Process finished with exit code 0
```

Question 5: Sum of Odd Numbers between 1 and N

Source code:

```java
package LB_Assingment4;

import java.util.Scanner;

public class Odd_no_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Number: ");
        int num=sc.nextInt();
        int add=0;
        for(int i=1;i<=num;i++){

            if(i%2!=0){
                add=add+i;
            }
        }
        System.out.println("The Sum of odd numbers from 1 to "+num+" is: "+add);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Number: 10
The Sum of odd numbers from 1 to 10 is: 25

Process finished with exit code 0
```

Question 6: Print All Elements of an Array

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Print_array_elements {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int [] arr=new int[5];
        System.out.print("Enter 5 Intergers: ");

        for(int i=0;i<arr.length;i++){
            arr[i]= sc.nextInt();
        }
        for(int num:arr){
            System.out.print(num+" ");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Intergers: 10 50 4 6 20
10 50 4 6 20
Process finished with exit code 0
```

Question 7: Find the Sum of All Elements in an Array

Source code:

```java
package LB_Assingment4;

import java.util.Scanner;

public class sum_of_arr_elements_foreach {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int []arr=new int[5];

        System.out.print("Enter 5 Integers: ");
        for(int i=0;i<arr.length;i++){
            arr[i]= sc.nextInt();
        }
        int add=0;
        for(int num:arr){
            add=add+num;
        }
        System.out.println("The sum of all numbers is: "+add);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Integers: 4 6 8 2 10
The sum of all numbers is: 30

Process finished with exit code 0
```

Question 8: Print All Names in a String Array

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class print_names_array {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        String [] arr=new String[4];
        System.out.println("Enter 4 names: ");
        for(int i=0;i<arr.length;i++){
            arr[i]=sc.nextLine();
        }
        for(String names:arr){
            System.out.print(names+" ");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 4 names: Pranav
manisha
sanjay
bagal
Pranav manisha sanjay bagal
```

Question 9: Find the Largest Element in an Array

Source code:

```
package LB_Assingment4;

import java.util.Arrays;
import java.util.Scanner;

public class largest_num_arr {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int []arr=new int[5];
        System.out.print("Enter 5 integers: ");

        for(int i=0;i< arr.length;i++){
            arr[i]=sc.nextInt();
        }
        Arrays.sort(arr);
        System.out.println("The Largest element is: "+arr[arr.length-1]);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 integers: 1 8 89 50 10
The Largest element is: 89

Process finished with exit code 0
```

Question 10: Find the Average of Elements in an Array

Source code:

```java
package LB_Assingment4;

import java.util.Scanner;

public class Avg_arr_elements {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int [] arr=new int[5];
        System.out.println("Enter 5 Numbers: ");
        for(int i=0;i< arr.length;i++){
            arr[i]=sc.nextInt();
        }
        int sum=0;
        for(int i=0;i<arr.length;i++){
            sum=sum+arr[i];
        }
        double avg=sum/ arr.length;
        System.out.println(avg);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Numbers:
10 20 30 40 50
30.0

Process finished with exit code 0
```

Question 11: Count Positive and Negative Numbers in an Array

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Count_positive_Negative_nos {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int [] arr=new int[5];
        System.out.println("Enter 5 Integers: ");

        for(int i=0;i< arr.length;i++){
            arr[i]=sc.nextInt();
        }
        int count_pos=0;
        int count_neg=0;
        for(int i=0;i< arr.length;i++){
            if(arr[i]>0){
                count_pos=count_pos+1;
            }else{
                count_neg=count_neg+1;
            }
        }
        System.out.println("Positive Numbers: "+count_pos);
        System.out.println("Negative Numbers: "+count_neg);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Integers:
-1 5 -8 3 2
Positive Numbers: 3
Negative Numbers: 2

Process finished with exit code 0
```

Question 12: Sort an Array in Ascending Order

Source code:

```
package LB_Assingment4;

import java.util.Arrays;
import java.util.Scanner;

public class Array_sorting {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int [] arr=new int[5];
        System.out.println("Enter 5 Integers: ");

        for(int i=0;i< arr.length;i++){
            arr[i]=sc.nextInt();
        }
        Arrays.sort(arr);
        System.out.print("Sorted Array: ");
        for(int num:arr){
            System.out.print(num+" ");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Integers:
5 3 4 1 2
Sorted Array: 1 2 3 4 5
Process finished with exit code 0
```

Question 13: Check if an Array Contains a Specific Element

Source code:

```
package LB_Assingment4;

import java.util.Arrays;
import java.util.Scanner;

public class Check_arr_contain_elements {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        Integer [] arr=new Integer[5];
        System.out.print("Enter 5 Integers: ");

        for(int i=0;i< arr.length;i++){
            arr[i]=sc.nextInt();
        }
        System.out.print("Enter the Number to Search: ");
        int find_num=sc.nextInt();
        if(Arrays.asList(arr).contains(find_num)){
            System.out.println("Found");
        }else{
            System.out.println("Not Found");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Integers: 10 20 30 40 50
Enter the Number to Search: 40
Found

Process finished with exit code 0
```

Question 14: Find the Index of an Element in an Array

Source code:

```
package LB_Assingment4;

import java.util.Arrays;
import java.util.Scanner;

public class Index_of_elem_arr {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int [] arr=new int[5];
        System.out.print("Enter 5 Integers: ");
        for(int i=0;i<arr.length;i++){
            arr[i]= sc.nextInt();
        }
        System.out.print("Enter the Number to search: ");
        int find_num= sc.nextInt();

        int index=Arrays.binarySearch(arr,find_num);
        if(index>=0){
            System.out.println("The number "+find_num+" is found at index "+index);
        }else{
            System.out.println("Not found");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Integers: 5 10 15 20 25
Enter the Number to search: 15
The number 15 is found at index 2


Process finished with exit code 0
```

Question 15:Write a program to print the following pattern:

```
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
```

Source code:

```
package LB_Assingment4;

public class Q15 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=i;j++){
                System.out.print(i);
                if(j<i){
                    System.out.print("*");
                }
            }
            System.out.println();
        }
        for(int i=5;i>=2;i--){
            for(int j=1;j<=i;j++){
                System.out.print(i);
                if(j<i){
                    System.out.print("*");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
```

Question 16: Write a program to print the following pattern:

```
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5
```

Source code:

```java
package LB_Assingment4;

public class Q16 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=i;j++){
                System.out.print(j);
                if(j<i){
                    System.out.print("*");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5

Process finished with exit code 0
```

Question 17: Write a program to print the following pattern:

```
1
1*3
1*3*5
1*3*5*7
1*3*5*7*9
```

Source code:

```
package LB_Assingment4;

public class Q17 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=2*i-1;j++){
                if(j%2!=0){
                    System.out.print(j);

                    if(j<2*i-1){
                        System.out.print("*");
                    }
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
1*3
1*3*5
1*3*5*7
1*3*5*7*9

Process finished with exit code 0
```

Question 18: Write a program to print the following pattern:

```
11111
22222
33333
44444
55555
```

Source code:

```java
package LB_Assingment4;

public class Q18 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=5;j++){
                System.out.print(i);
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-
11111
22222
33333
44444
55555

Process finished with exit code 0
```

Question 19: Write a program to print the following pattern:

```
1
22
333
4444
55555
```

Source code:

```
package LB_Assingment4;

public class Q19 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=i;j++){
                System.out.print(i);
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
22
333
4444
55555

Process finished with exit code 0
```

Question 20: Write a program to print the following pattern

```
1
12
123
1234
12345
```

Source code:

```
package LB_Assingment4;

public class Q20 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=i;j++){
                System.out.print(j);
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
12
123
1234
12345

Process finished with exit code 0
```

Question 21: Write a program to print the following pattern:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Source code:

```
package LB_Assingment4;

public class Q21 {
    public static void main(String[] args) {
        int num=1;
        for(int i=1;i<=5;i++){
            for(int j=1;j<=i;j++){
                System.out.print(num+" ");
                num++;
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

Process finished with exit code 0
```

Question 22: Write a program to print the following pattern:

```
******
*    *
*    *
*    *
*    *
******
```

Source code:

```
package LB_Assingment4;

public class Q22 {
    public static void main(String[] args) {
        for(int i=1;i<=6;i++){
            for(int j=1;j<=6;j++){
                if(i==1 || i==6 || j==1 || j==6){
                    System.out.print("*");
                }else{
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
******
*    *
*    *
*    *
*    *
******

Process finished with exit code 0
```

Question 23: Write a program to print the following pattern:

```
     *
    ***
   *****
  *******
 *********
***********
 *********
  *******
   *****
    ***
     *
```

Source code:

```
package LB_Assingment4;

public class Q23 {
    public static void main(String[] args) {
        for(int i=1;i<=7;i++){
            for(int j=1;j<=7-i;j++){
                System.out.print(" ");
            }
            for(int j=1;j<=2*i-1;j++){
                System.out.print("*");
            }
            System.out.println();
        }
        for(int i=6;i>=1;i--){
            for(int j=1;j<=7-i;j++){
                System.out.print(" ");
            }
            for(int j=1;j<=2*i-1;j++){
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Output:

Question 24: Reverse a String

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Q24 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a String: ");
        String word=sc.nextLine();

        System.out.print("Reversed String: ");
        for(int i=word.length()-1;i>=0;i--){
            System.out.print(word.charAt(i));
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe" '
Enter a String: Pranav
Reversed String: vanarP
Process finished with exit code 0
```

Question 25: Count Vowels in a String

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Q25 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a String: ");
        String word=sc.nextLine();

        int count=0;
        for(int i=0;i<word.length();i++){
            char ch=word.charAt(i);
            if(ch == 'a' ||ch == 'e' ||ch == 'i' ||ch == 'o' ||ch == 'u'
                    ||ch == 'A' ||ch == 'E' ||ch == 'I' ||ch == 'O' ||ch == 'U'){
                count++;
            }
        }
        System.out.println("The number of vowels in '" + word + "' is: " + count);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a String: Rainbow
The number of vowels in 'Rainbow' is: 3

Process finished with exit code 0
```

Question 26: Check if a String is a Palindrome

Source code:

```java
package LB_Assingment4;

import java.util.Scanner;

public class Q26 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a String: ");
        String word = sc.nextLine();

        String rev = "";
        for (int i=word.length()-1;i>=0;i--){
            rev =rev+word.charAt(i);
        }
        if(word.equals(rev)){
            System.out.println("The String '"+word+"' is a palindrome");
        }else{
            System.out.println("The String '"+word+"' is not a palindrome");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a String: kanak
The String 'kanak' is a palindrome

Process finished with exit code 0
```

Question 27: String Literal and Object Creation

Source code:

```
package LB_Assingment4;

public class Q27 {
    public static void main(String[] args) {
        String str1="hello";
        String str2="hello";

        if(str1==str2){
            System.out.println("Both variables point to the same object: "+(str1==str2));
        }else {
            System.out.println("Both variables point to the same object: "+(str1==str2));
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-jav
Both variables point to the same object: true

Process finished with exit code 0
```

Question 28: String Creation with new Keyword

Source code:

```
package LB_Assingment4;

public class Q28 {
    public static void main(String[] args) {
        String str1=new String("Hello");
        String str2=new String("Hello");

        System.out.println("Using ==: "+(str1==str2));
        System.out.println("Using .equals(): "+(str1.equals(str2)));
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Using ==: false
Using .equals(): true

Process finished with exit code 0
```

Question 29: String Concatenation and Object Creation

Source code:

```
package LB_Assingment4;

public class Q29 {
    public static void main(String[] args) {
        String str1="Hello";
        String str2="World";
        String str3=str1+str2;

        System.out.println("Is str3 pointing to the same object as str1? "+(str1==str3));
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-javaagent:C
Is str3 pointing to the same object as str1? false

Process finished with exit code 0
```

Question 30: String Pool with intern() Method

Source code:

```
package LB_Assingment4;

public class Q30 {
    public static void main(String[] args) {
        String str1=new String("Hello");
        String str2=str1.intern();
        String str3="Hello";

        System.out.println("Is str2 and str3 pointing to the same object? "+(str2==str3));
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-javaagent:C
Is str2 and str3 pointing to the same object? true

Process finished with exit code 0
```

Question 31: Multiple String Literals with Same Content

Source code:

```
package LB_Assingment4;

public class Q31 {
    public static void main(String[] args) {
        String str1="Java";
        String str2="Java";
        String str3="Java";

        System.out.println("All String point to the same object: "+(str1==str2 && str2==str3));
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-javaagent:C:\P
All String point to the same object: true

Process finished with exit code 0
```